Smart Roadmap Builder

The Smart Roadmap Builder is a web-based system designed to generate personalized learning roadmaps based on user goals, daily availability, and preferences. By transforming complex learning objectives into structured, step-by-step daily tasks, the system effectively mitigates procrastination, reduces overwhelm, and significantly boosts learner productivity.

PROJECT TYPE	PLATFORM	PRIMARY USERS	CORE VALUE
Web Application	**Responsive Web**	**Multi Stakeholders**	**Structured Task Planning**

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1. Project Planning & Management

A. Project Proposal

Overview: The system provides automated micro-planning for learners, directly addressing the difficulty of navigating unstructured educational materials.

Problem Statement: Learners frequently face cognitive overload, lack of structured paths, and procrastination due to overwhelming resources and poorly defined milestones.

SMART Objectives

Specific

Develop a web platform that parses user goals

Measurable

Achieve a >90% task generation accuracy and 80% user task completion rate.

Achievabl

Utilize proven modern web stacks (React,

Relevant

Directly solves the identified issues of procrastination and unstructured study.

Time-bound

Deliver a functional, tested prototype by the

IN-SCOPE

- User Registration & Secure Authentication
- Goal Definition & Parameter Input (Time, Preference)
- Automated Roadmap Generation Engine

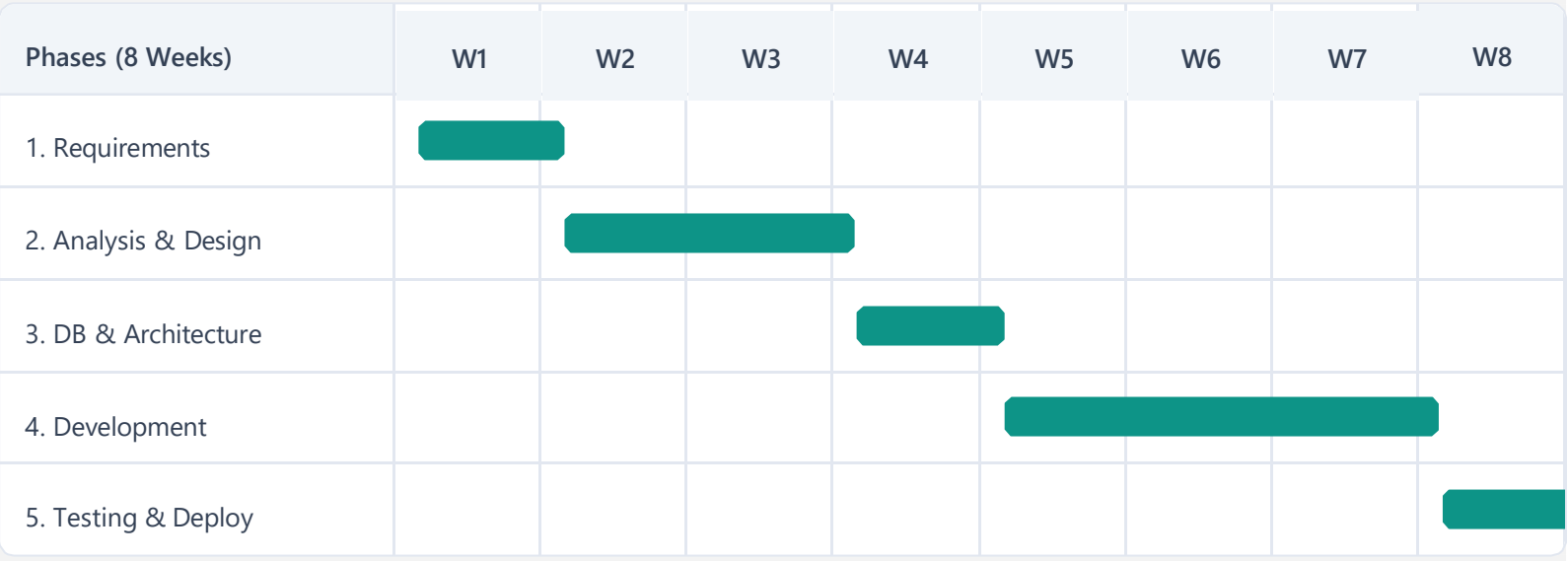
OUT-OF-SCOPE

- Direct Content Hosting or Video Streaming
- Social Networking & Community Forums
- Native iOS/Android Applications
- Monetization & Payment Gateways

- Daily Task Interaction & Progress Tracking
- Visual Dashboard for Progress Overview

B. Project Plan

Phase	Deliverables
1. Initiation & Requirements	Requirements Document, Project Proposal, Stakeholder Analysis
2. System Analysis & UI/UX	Use Case Models, Wireframes, Figma Prototypes, UI Guidelines
3. Architecture & DB Design	ERD, Database Schema, System Architecture Diagram, API Specs
4. Frontend & Backend Dev	Functional Web App, RESTful APIs, Integrated Frontend/Backend
5. Testing, QA & Deployment	Test Cases, Bug Reports, Deployed Application, Final Presentation



C. Task Assignment & Roles

Frontend Developer

Builds UI, state management, API integration.

Backend Developer

Builds REST APIs, auth logic, generation algorithms.

UI/UX Designer

Creates wireframes, high-fidelity mockups, user flows.

Database Engineer

Designs schema, ensures data integrity and indexing.

Project Manager

Oversees schedule, risk management, and documentation.

D. Risk Assessment & Mitigation

Risk	Description	Impact	Probability	Mitigation Strategy
Roadmap Accuracy Risk	Generated tasks do not logically align with the user's overarching goals.	HIGH	Medium	Implement robust rule-based parsing and allow manual user task adjustments.
User Retention Risk	Users abandon the system after generating their first roadmap.	HIGH	Medium	Integrate gamification elements and daily reminder notifications.
Performance Issues	Slow generation times leading to user frustration.	MEDIUM	Low	Optimize backend algorithms and employ database indexing.
Security Vulnerabilities	Unauthorized access to user data or injection attacks.	HIGH	Low	Use JWT, bcrypt for passwords, and sanitize all inputs.

Scope Creep	Unplanned features delaying core academic deliverables.	MEDIUM	High	Strict adherence to "In-Scope" features and rigorous PM oversight.

Risk	Description	Impact	Probability	Mitigation Strategy
Technical Complexity	Integration challenges between frontend and backend architectures.	LOW	Medium	Establish clear API contracts and conduct frequent integration testing.

Risk Summary Note: Proactive mitigation focuses heavily on algorithm accuracy and scope containment to ensure delivery within the academic timeframe.

E. KPIs (Key Performance Indicators)

KPI	Target	Why It Matters	Measurement Method
System response time	< 2 seconds	Ensures a fluid, responsive user experience.	Server logs, APM Tools
Roadmap generation time	< 2 seconds	Prevents user drop-off during the critical setup phase.	Backend performance profiling
User retention rate	> 60% (Week 1)	Validates the usefulness of the structured planning.	Database login records
Task completion rate per user	> 75%	Indicates the roadmap chunks are realistic and actionable.	Task status queries
System uptime	≥ 99%	Ensures reliability for daily user interaction.	Uptime monitoring software
User engagement rate	Daily logins	Measures the success of the habit-building aspect of the tool.	Session tracking

2. Literature Review & Academic Evaluation

In educational technology, personalized learning and micro-planning are paramount. Research demonstrates that breaking complex, long-term educational objectives into granular, daily tasks minimizes cognitive load and drastically reduces procrastination. The Smart Roadmap Builder synthesizes these productivity systems by enforcing daily structured task planning, transforming overwhelming resources into actionable, bite-sized daily achievements.

Criteria	Description
Functionality	Core roadmap generation and progress tracking work flawlessly without critical errors.
Innovation	Unique approach to transforming static goals into dynamic daily tasks.
Usability	Interface is intuitive, minimal, and adheres to modern UX principles.
Documentation Quality	Comprehensive, academic-grade reports, diagrams, and code comments.
Implementation Quality	Clean, modular architecture using industry-standard best practices.

Suggested Improvements

AI-based Adaptive Generation

Integrate LLMs to generate highly specific niche roadmaps dynamically.

Personalized Optimization

Adjust future task loads automatically based on past completion speeds.

Mobile App Extension

Develop React Native clients for on-the-go notifications and tracking.

Advanced Analytics

Provide detailed charts showing learning velocity and time-spent metrics.

Gamification

Introduce streaks, badges, and points to further enhance retention.

Final Grading Criteria

Component	Percentage Weight
Documentation	25%
Implementation	35%
Testing	15%
UI/UX Design	15%
Presentation	10%

3. Requirements Gathering

A. Stakeholder Analysis

Stakeholder	Role	Needs / Expectations
End Users	Primary consumers	Intuitive interface, accurate roadmap generation, reliable tracking.
Developers	System creators	Clear technical requirements, solid architecture, realistic deadlines.
Academic Supervisor	Evaluator	Adherence to SE methodologies, clean documentation, functional prototype.

B. User Stories

1. *"As a user, I want to securely log in so that my data and progress are saved."*

2. *"As a user, I want to input my learning goal and available daily time so that the system understands my constraints."*

3. *"As a user, I want the system to generate a day-by-day roadmap so that I know exactly what to study."*

4. *"As a user, I want to mark daily tasks as complete so that I can track my overall progress."*

5. *"As a user, I want to view a dashboard of my progress so that I stay motivated."*

6. *"As a user, I want a clean, minimal interface so that I am not distracted from my actual learning."*

C. Use Cases

Use Case	Primary Actor	Preconditions	Main Flow	Outcome
Login	User	Has registered account	Enters credentials → System validates → Grants access	User is authenticated and routed to dashboard
Goal Setting	User	Authenticated	Selects topic → Inputs time constraints → Submits	System stores goal parameters
Roadmap Generation	System	Goal parameters exist	Processes inputs → Breaks down into tasks → Saves to DB	A sequential task list is presented to the user
Daily Task Interaction	User	Roadmap exists	Views today's tasks → Completes study → Checks off task	Task status updated to completed
Progress Tracking	User	Has interacted with tasks	Navigates to dashboard → Views completion charts	User understands current standing

D. Functional Requirements

Req ID	Formal Requirement Statement
FR-01	The system shall provide authentication and user registration capabilities.
FR-02	The system shall allow users to define structured learning goals and time parameters.

Req ID	Formal Requirement Statement
FR-03	The system shall generate personalized roadmaps based on user inputs.
FR-04	The system shall break broad goals into discrete, daily executable tasks.
FR-05	The system shall track user progress and calculate completion percentages.
FR-06	The system shall store user data securely within a relational database.
FR-07	The system shall provide a responsive UI compatible with desktop and mobile viewports.

E. Non-Functional Requirements

Performance

Fast response time (< 2 seconds). Efficient and optimized roadmap generation algorithms.

Security

JWT-based authentication system. Encrypted data storage (passwords). Secure API communication via HTTPS.

Usability

Simple, minimal, distraction-free interface. One-task focus design paradigm. Fully responsive design.

Reliability

99% system uptime requirement. Auto-save functionality. Robust error handling preventing system crashes.

4.Design Requirements & Diagrams

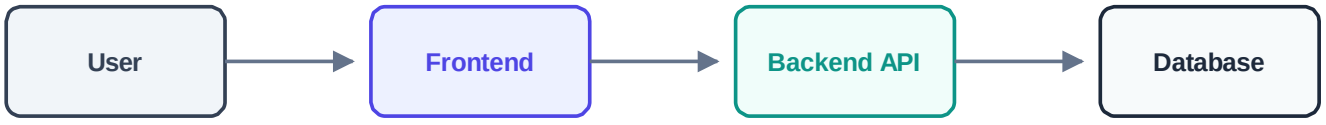


Figure 1: High-Level System Architecture Diagram

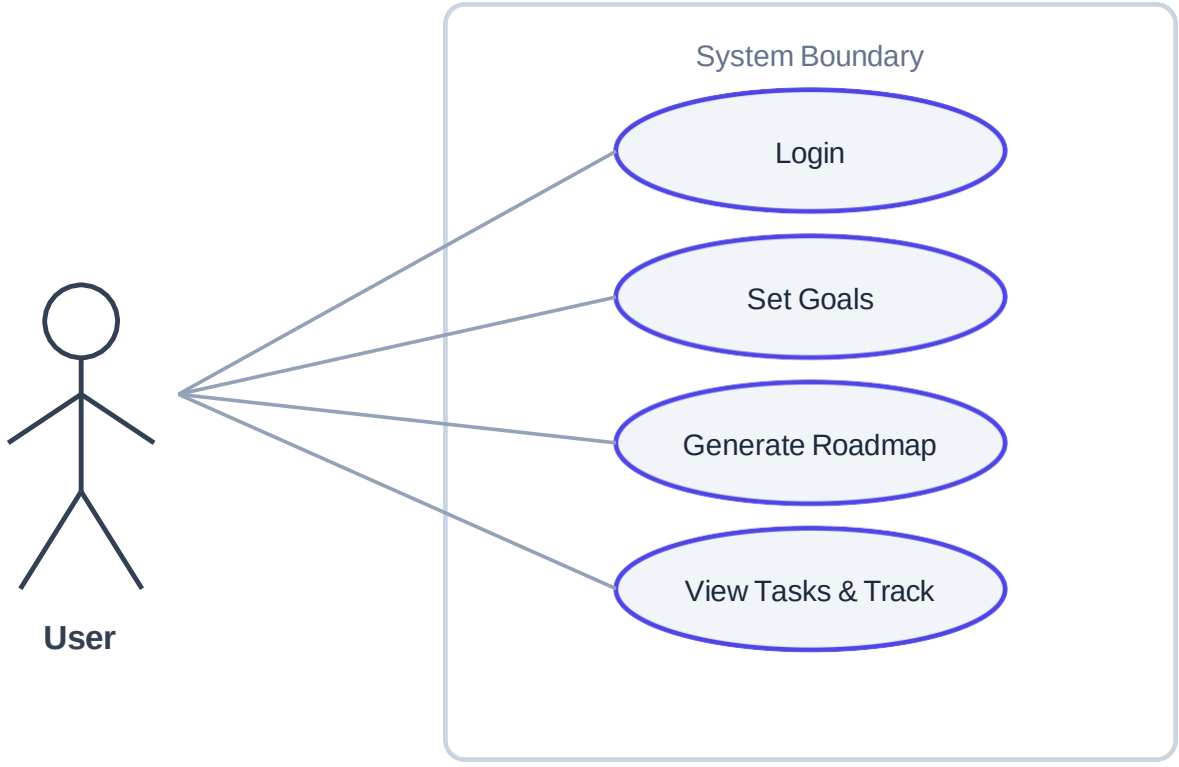


Figure 2: Use Case Diagram

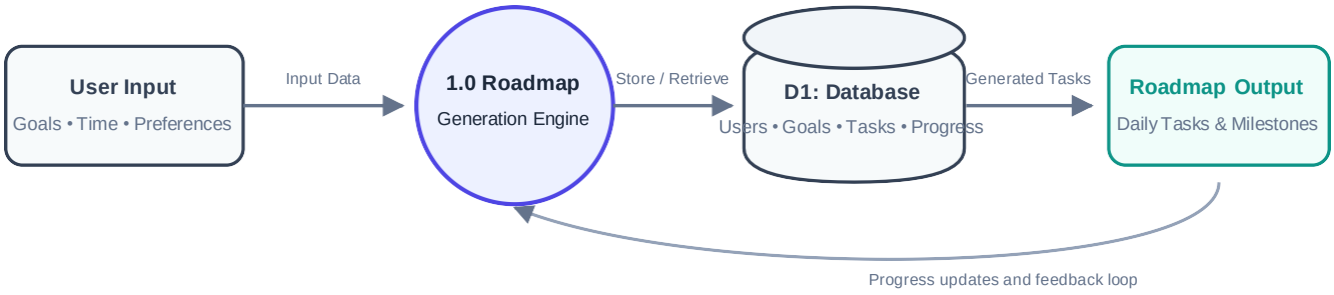


Figure 3: Data Flow Diagram (Level 0)

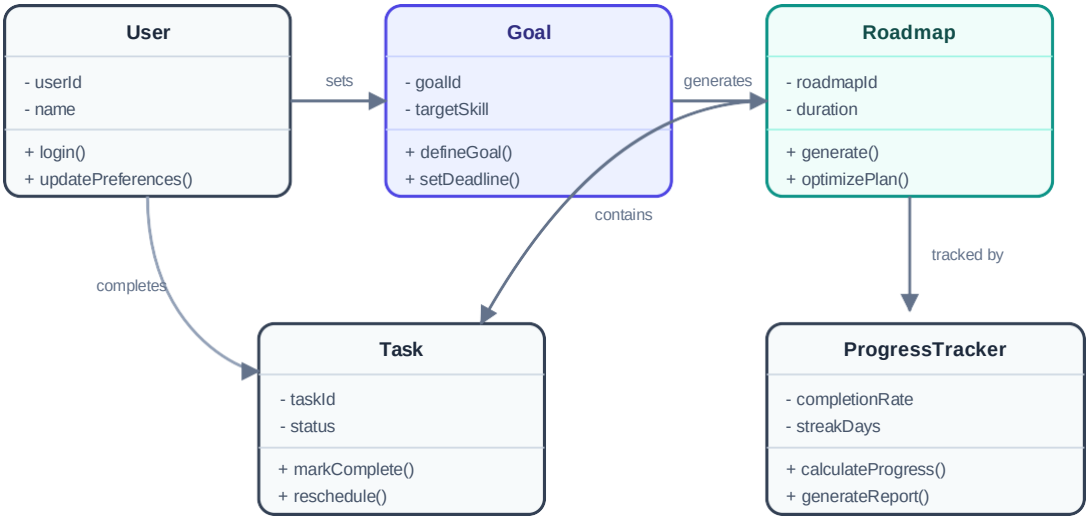


Figure 4: Simplified Class Diagram for Core Domain Entities

5.Conclusion

The Smart Roadmap Builder is a robust, practical, and scalable web application aimed at modernizing self-directed education. By incorporating strict software engineering practices—comprehensive requirement elicitation, structured architectural design, and thorough risk mitigation—the project stands as a high-value tool for enhancing productivity and achieving long-term learning goals incrementally.